Local Notifications

1. Setup

Dependencies

1. To use the flutter_local_notifications package, add this to your pubspec.yaml

```
flutter_local_notifications: ^19.1.0
timezone: ^0.10.1
```

This one is is for scheduled notifications.

2. Run this in your terminal

```
flutter pub get
```

Android permissions

1. To push notifications in your Android app, you need to set up permissions first. Add these to your app/src/main/AndroidManifest.xml.

```
<uses-permission android:name="android.permission.POST_NOTIFICATIONS"/>
```

POST_NOTIFICATIONS: Allows the app to show notifications (Android 13+).

```
<uses-permission
android:name="android.permission.RECEIVE_BOOT_COMPLETED"/>
```

RECEIVE_BOOT_COMPLETED: Lets the app run code after the device restarts.

```
<uses-permission android:name="android.permission.VIBRATE"/>
```

VIBRATE: Enables the app to make the device vibrate.

```
<uses-permission
android:name="android.permission.ACCESS_NOTIFICATION_POLICY" />
```

ACCESS_NOTIFICATION_POLICY: Lets the app read or change Do Not Disturb settings.

This is how it should look like.

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.POST_NOTIFICATIONS"/>
    <uses-permission android:name="android.permission.RECEIVE_BOOT_COMPLETED"/>
    <uses-permission android:name="android.permission.VIBRATE"/>
    <uses-permission android:name="android.permission.ACCESS_NOTIFICATION_POLICY" />

    <application
        android:label="local_notifications"
        android:name="${applicationName}"
```

2. Now you need to add receivers for the notifications to work all the time.

```
<receiver
android:exported="false"
android:name="com.dexterous.flutterlocalnotifications.ScheduledNotificationReceiver"
/>
```

ScheduledNotificationReceiver: Handles the triggering of scheduled notifications

```
<receiver
android:exported="false"
android:name="com.dexterous.flutterlocalnotifications.ScheduledNotificationBootReceiver">
<intent-filter>
<action android:name="android.intent.action.BOOT_COMPLETED"/>
<action android:name="android.intent.action.MY_PACKAGE_REPLACED"/>
<action android:name="android.intent.action.QUICKBOOT_POWERON" />
<action android:name="com.htc.intent.action.QUICKBOOT_POWERON"/>
</intent-filter>
</receiver>
```

ScheduledNotificationBootReceiver: Reschedules notifications after reboot/App reinstall/update & Certain manufacturer-specific quick boot

```
<receiver
android:exported="false"
android:name="com.dexterous.flutterlocalnotifications.FlutterLocalNotificationsReceiver"/>
```

FlutterLocalNotificationsReceiver: Showing notifications immediately

This is how it should look like

```
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>

    <!-- Handles the triggering of scheduled notifications -->
    <receiver
        android:exported="false"
        android:name="com.dexterous.flutterlocalnotifications.ScheduledNotificationReceiver" />

    <!-- Reschedules notifications after reboot/App reinstall/update & Certain manufacturer-specific quick boot -->
    <receiver
        android:exported="false"
        android:name="com.dexterous.flutterlocalnotifications.ScheduledNotificationBootReceiver">
        <intent-filter>
            <action android:name="android.intent.action.BOOT_COMPLETED"/>
            <action android:name="android.intent.action.MY_PACKAGE_REPLACED"/>
            <action android:name="android.intent.action.QUICKBOOT_POWERON" />
            <action android:name="com.htc.intent.action.QUICKBOOT_POWERON"/>
        </intent-filter>
    </receiver>

    <!-- Showing notifications immediately -->
    <receiver
        android:exported="false"
        android:name="com.dexterous.flutterlocalnotifications.FlutterLocalNotificationsReceiver"/>

    <!-- Don't delete the meta-data below.
         This is used by the Flutter tool to generate GeneratedPluginRegistrant.java -->
```

IOS permissions

1. To push notifications in your IOS app, you need to set up permissions first. Add these to your ios/Runner/AppDelegate.swift.

```
import flutter_local_notifications
FlutterLocalNotificationsPlugin.setPluginRegistrantCallback { (registry) in
GeneratedPluginRegistrant.register(with: registry)
 }
if #available(iOS 10.0, *) {
UNUserNotificationCenter.current().delegate = self as?
UNUserNotificationCenterDelegate
}
```

What is this? I don't know, i'm not a Swift dev, but in the package documentation, they ask us to add this.

This is how it should look like

```
import Flutter
import UIKit
import flutter_local_notifications //new

@main
@objc class AppDelegate: FlutterAppDelegate {
  override func application(
    _ application: UIApplication,
    didFinishLaunchingWithOptions launchOptions: [UIApplication.LaunchOptionsKey: Any]?
  ) -> Bool {
    //new
    FlutterLocalNotificationsPlugin.setPluginRegistrantCallback { (registry) in
    GeneratedPluginRegistrant.register(with: registry)
    }
    GeneratedPluginRegistrant.register(with: self)
    //new
    if #available(iOS 10.0, *) {
      UNUserNotificationCenter.current().delegate = self as? UNUserNotificationCenterDelegate
    }
    return super.application(application, didFinishLaunchingWithOptions: launchOptions)
  }
}
```

Pro tip: Flutter Clean

If you face any problem, clean your app:

```
flutter clean

flutter pub get
```

If you are on mac, you can run this also.

```
cd ios

pod install

cd ..
```

2. Code

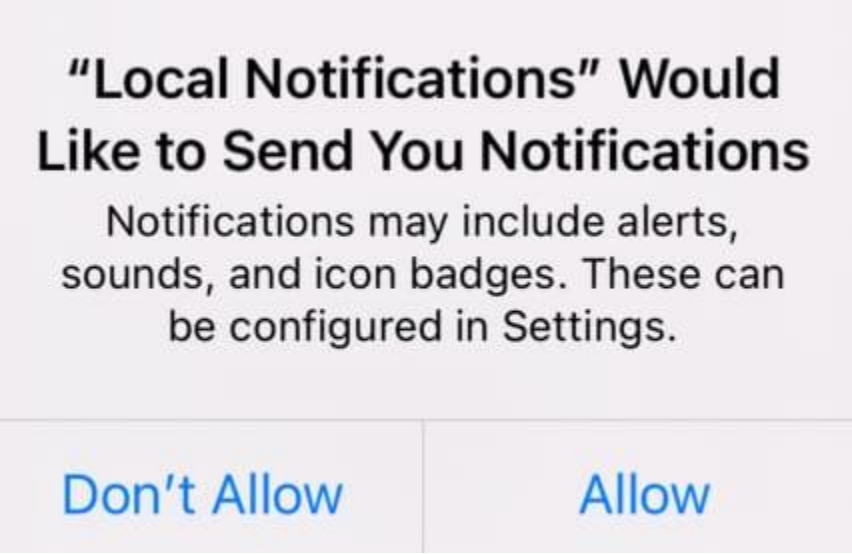
Init

1. Let's initialize the notifications for android and ios.

```
Future<void> init() async {
  initializeTimeZones();

  //! https://en.wikipedia.org/wiki/List_of_tz_database_time_zones
  setLocalLocation(
    getLocation('America/Toronto'),
  );

  const androidSettings =
      AndroidInitializationSettings('@mipmap/launcher_icon');
  const DarwinInitializationSettings iosSettings =
      DarwinInitializationSettings();

  const InitializationSettings initializationSettings =
      InitializationSettings(
    android: androidSettings,
    iOS: iosSettings,
  );
  await notificationsPlugin.initialize(initializationSettings);
}
```

Visit this Wikipedia page to find your location.

2. Now when you launch your application, you should see this if you run the future init().



3. Allow the notification otherwise it won't work.

Instant notifications

```
Future<void> showInstantNotification({
  required int id,
  required String title,
  required String body,
}) async {
  await notificationsPlugin.show(
    id,
    title,
    body,
    const NotificationDetails(
      android: AndroidNotificationDetails(
        'instant_notification_channel_id',
        'Instant Notifications',
        channelDescription: 'Instant notification channel',
        importance: Importance.max,
        priority: Priority.high,
      ), // AndroidNotificationDetails
      iOS: DarwinNotificationDetails(),
    ), // NotificationDetails
  );
}
```

Scheduled notifications

```dart
Future<void> scheduleReminder({
  required int id,
  required String title,
  String? body,
}) async {
  TZDateTime now = TZDateTime.now(local);
  TZDateTime scheduledDate = now.add(
    Duration(seconds: 3),
  );

  await notificationsPlugin.zonedSchedule(
    id,
    title,
    body,
    scheduledDate,
    const NotificationDetails(
      android: AndroidNotificationDetails(
        'daily_reminder_channel_id', // A unique ID to group notifications together.
        'Daily Reminders', // A human-readable name shown to users in their notification settings.
        channelDescription: 'Reminder to complete daily habits',
        importance: Importance.max,
        priority: Priority.high,
      ), // AndroidNotificationDetails
      iOS: DarwinNotificationDetails(),
    ), // NotificationDetails
    androidScheduleMode: AndroidScheduleMode.inexactAllowWhileIdle,
    matchDateTimeComponents:
        DateTimeComponents.dayOfWeekAndTime, // or dateAndTime
  );
}
```

3. Above an beyond

1. Get local timezone automatically

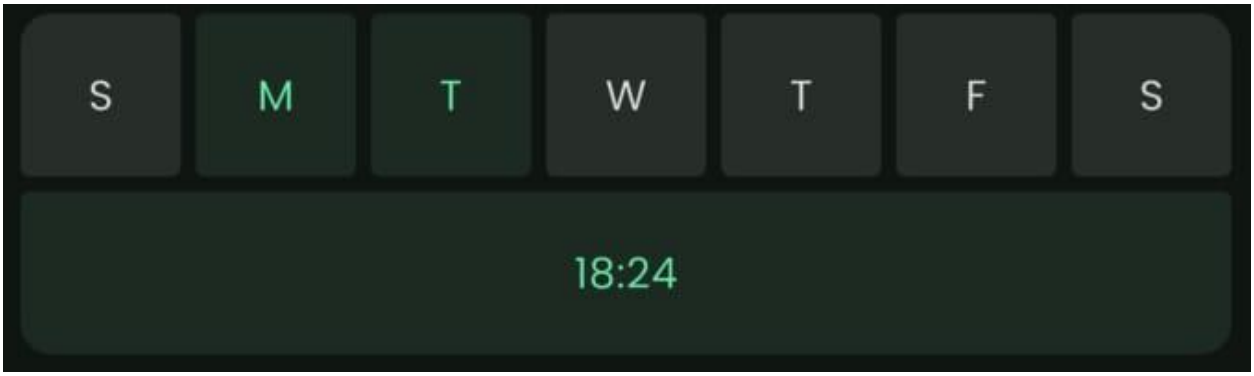
```dart
try {
  final String currentTimeZone = await FlutterTimezone.getLocalTimezone();
  tz.setLocalLocation(tz.getLocation(currentTimeZone));
} catch (e) {
  tz.setLocalLocation(tz.getLocation('America/Toronto')); // fallback
}
```

Pro tip: Each scheduled notification need a different ID!

2. Notification per day

Pro tip: Each scheduled notification need a different ID!

In this example, only Monday and Tuesday at 18:24 will trigger notifications

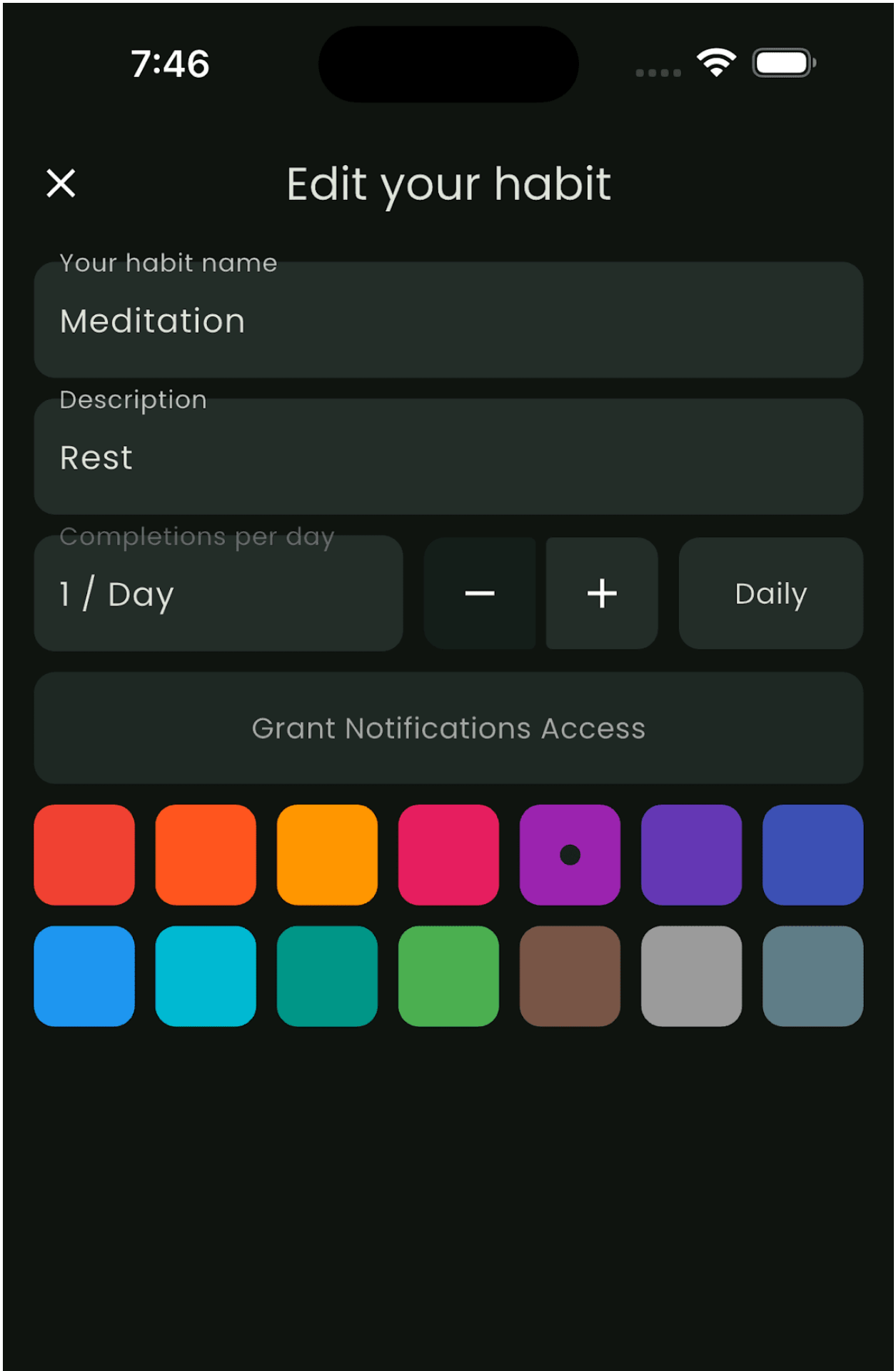


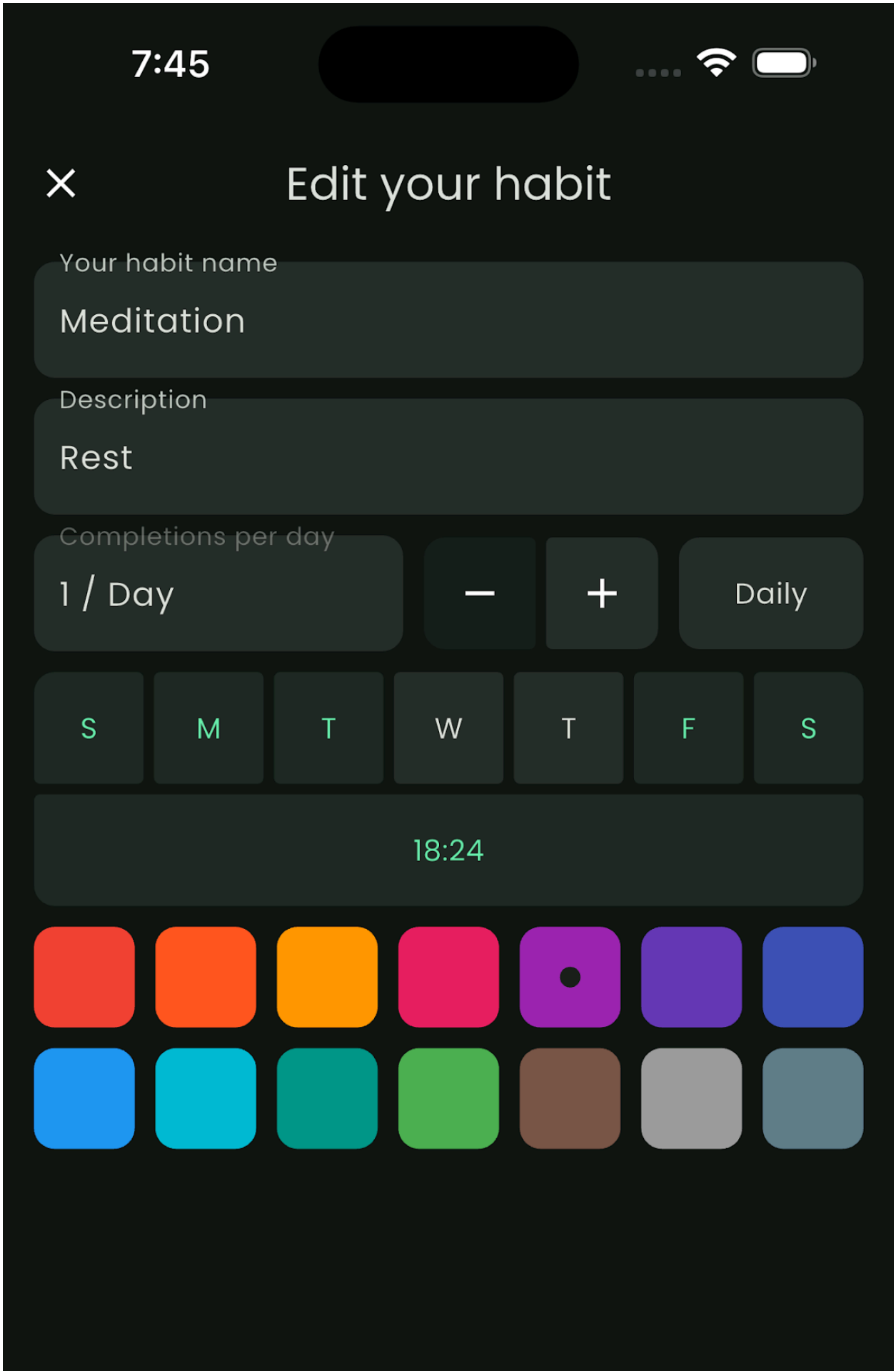
3. Grant notification access

If the user has rejected the local notifications popup 'Don't allow', then you can use the app_settings package to point the user into the notification settings.

```
app_settings: ^6.1.1
```

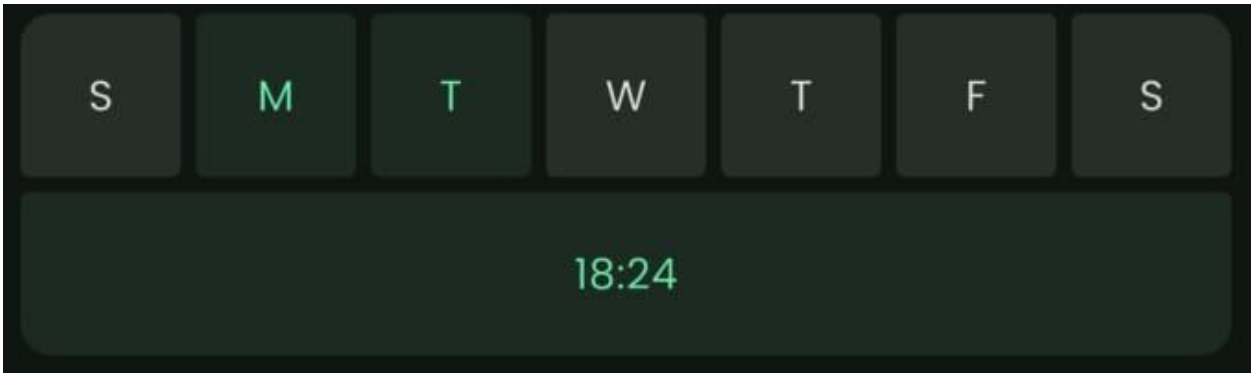
Then you create a logic to show the option only if the user has granted the notifications like the images down bellow.


7:46
Edit your habit
Your habit name
Meditation
Description
Rest
Completions per day
1 / Day
Daily
Grant Notifications Access


7:45
Edit your habit
Your habit name
Meditation
Description
Rest
Completions per day
1 / Day
Daily
S
M
T
W
T
F
S
18:24

4. Show pending saved notifications

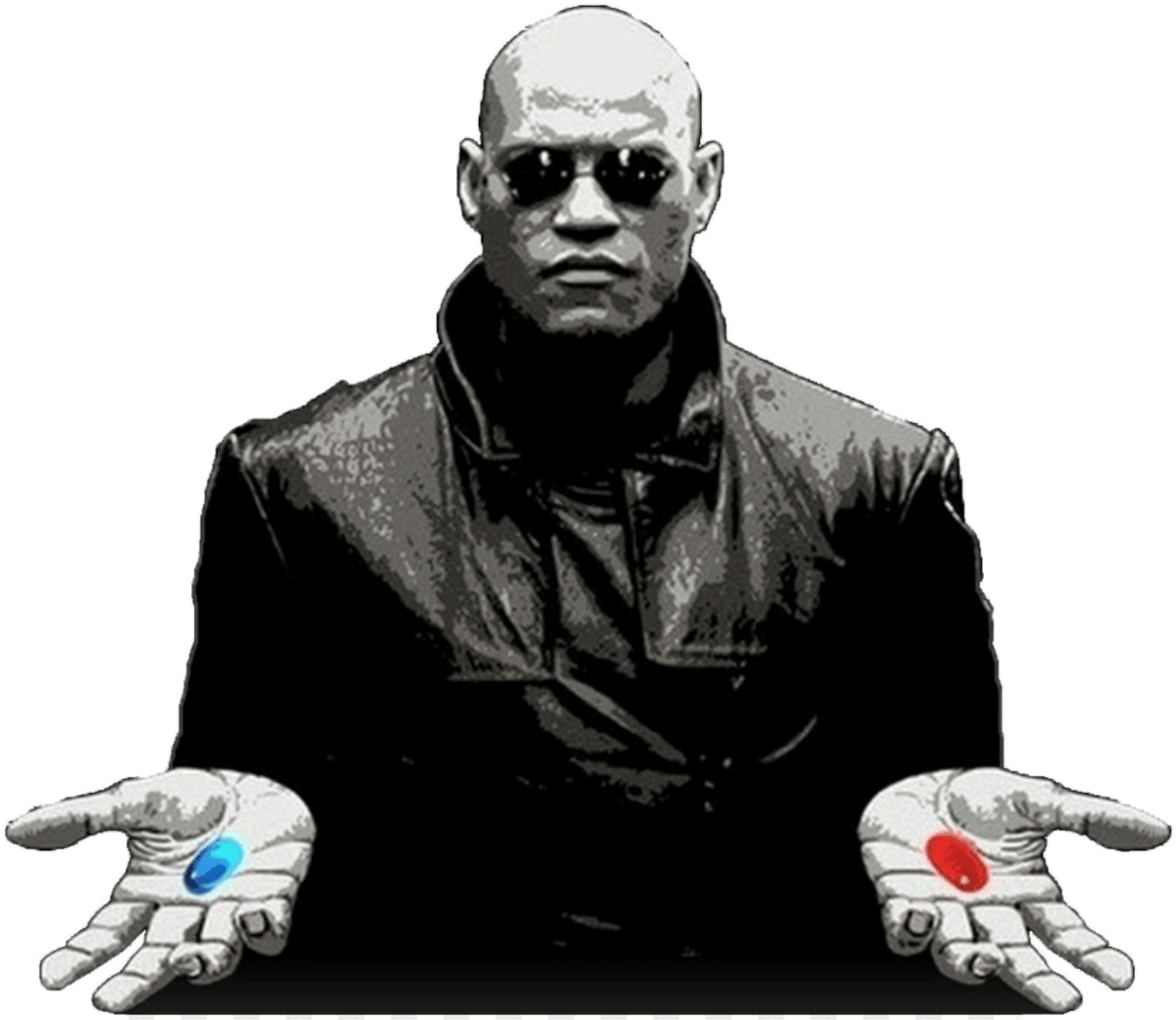
This information is saved locally into the phone, so we know that there is 2 schedules notifications for Monday and Tuesday. Remember, each notification need a unique ID.



5. Finally you will need to cancel reminders (scheduled notifications)

Use the cancelAll() and the cancel(id) of the flutter_local_notifications package to delete either all or specific scheduled notification.

```
.cancelAll()
.cancel(id)
```



PS: I would take both pills.